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Computational resources in the management of antibiotic resistance: Speeding up drug discovery

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This article reviews more than 50 computational resources developed in past two decades for forecasting of antibiotic resistance (AR)-associated mutations, genes and genomes. More than 30 databases have been developed for AR-associated information, but only a fraction of them are updated regularly. A large number of methods have been developed to find AR genes, mutations and genomes, with most of them based on similarity-search tools such as BLAST and HMMER. In addition, methods have been developed to predict the inhibition potential of antibiotics against a bacterial strain from the whole-genome data of bacteria. This review also discuss computational resources that can be used to manage the treatment of AR-associated diseases.

Keywords: Antibiotic resistance; Computational biology; Databases; In silico tools

Background

Antibiotic resistance (AR) is the capability of bacteria to resist medication that once could inhibit them. Bacteria undergo adaptive changes in response to the prescribed medication, resulting in the survival of bacteria in treated patients. Multidrug resistance (MDR) is simultaneous resistance against multiple antibiotics and is emerging as a major global threat. This is attributed to the misuse and overuse of medication and the reduced discovery of new drugs. There is a strong correlation between the use of an antibiotic in the treatment of bacterial disease and the subsequent development of antibiotic resistance. Bacteria develop multiple mechanisms of resistance when under selective pressure resulting from the overuse of antibiotics [1]. This phenomenon facilitates the growth of resistant bacteria and the death of vulnerable ones.

The mechanism of resistance could be either intrinsic or acquired. An intrinsic mechanism is attained due to the selection of chromosomal genes, whereas an acquired mechanism relies on horizontal gene transfer (HGT) [2,3]. In intrinsic mechanisms, resistance develops mainly through: (i) modification of the target of an antimicrobial agent that reduces the affinity of the target for the agents; (ii) altered porins, resulting in a change in the permeability of the bacterial outer membrane; (iii) increased expression of efflux pumps, causing the extrusion of antibiotics; and (iv) inactivation of the antibiotic, by its degradation or modification (Fig. 1). In an acquired mechanism, resistance develops through the acquisition of foreign DNA coding for resistance through HGT [4]. HGT leads to the conversion of commensal and environmental bacterial species into pathogenic ones. HGT occurs by one of three mechanisms, conjugation, transformation, and transduction, among which conjugation is thought to have the greatest influence [5].

The CTX-M family of genes, which confer resistance to cephalosporin group of antibiotics, have made the treatment of serious nosocomial infections a difficult challenge [6]. These genes first originated in the chromosomal DNA of various environmental *Kluyvera* species, but they have now spread to different bacterial species because they provide various resistance mechanisms. Similarly, OXA-48 genes, which are responsible for resistance to the carbapenem group of antibiotics, initially originated in

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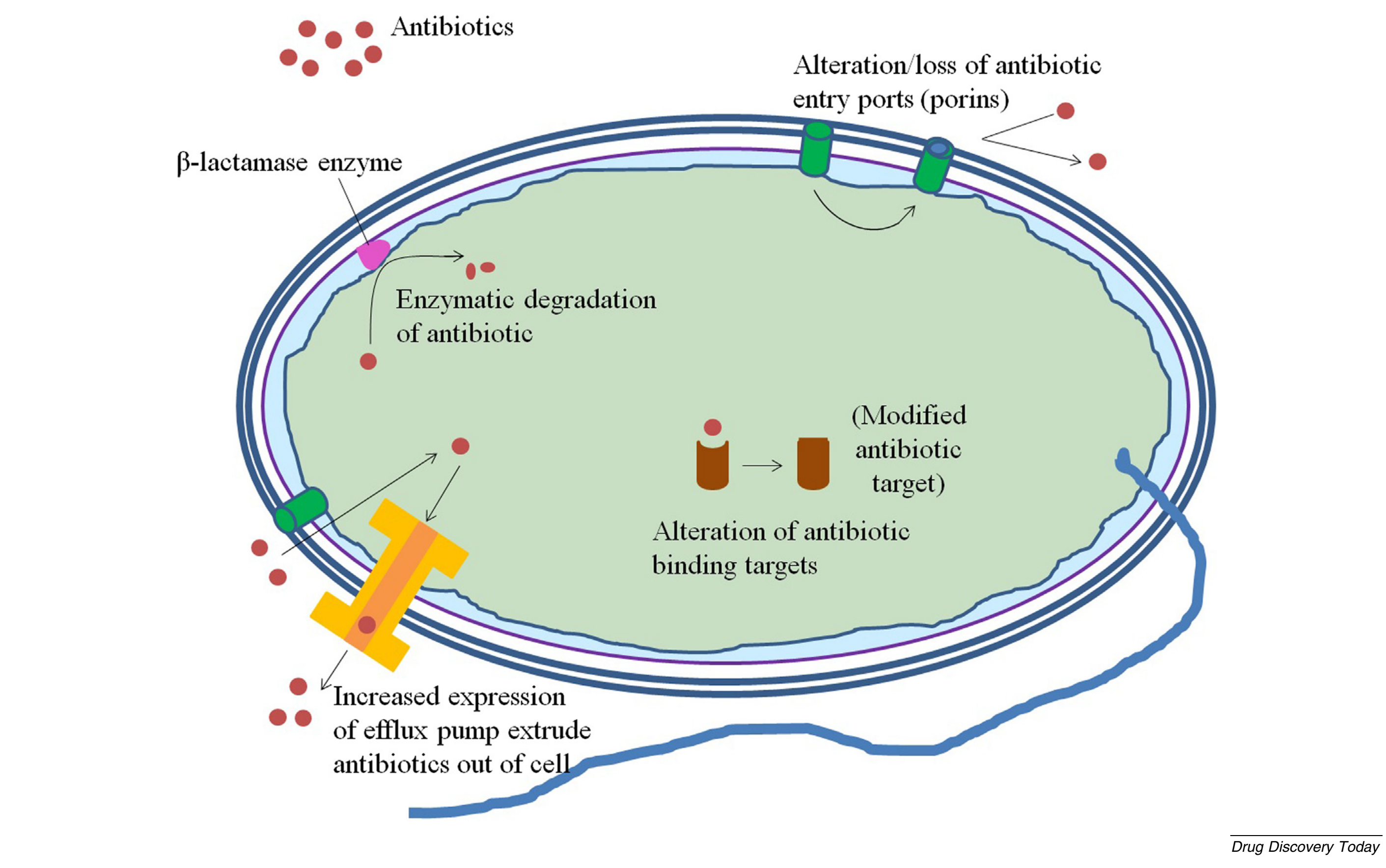

FIG. 1
Mechanisms of antibiotic resistance.

Shewanella species. Now, these genes are found in various enterobacterial species worldwide. Another plasmid-encoded qnrA gene, which plays a role in conferring resistance to quinolone antibiotic groups, was initially found in a reservoir of different Vibrionaceae species, but has now disseminated to various *Enterobacteriaceae* species globally [5]. The causative agent of tuberculosis (TB), *Mycobacterium tuberculosis*, has acquired resistance to multiple antibiotics. This occurs because the bacteria use a series of intrinsic resistance mechanisms to allow the active neutralization of antibiotic action [7]. Many other clinically relevant resistance genes have potentially originated from non-pathogenic bacteria, demonstrating the immense capability of bacteria to use horizontal gene transfer.

The property of resistance is encoded by naturally occurring genes that are present on bacterial chromosomes referred to as resistance markers. Numerous resistance markers are present in bacterial populations, and each type of marker has several variants. Moreover, newer variants are continuously emerging due to mutations and are disseminating throughout the world. Resistance markers can also move within or between genomes through transposons, integrons, or plasmids, and in doing so, could convert a susceptible bacterium into a resistant one. These resistant bacteria have the potential to cause serious medical crises and infection outbreaks.

The presence of multi-antibiotic resistant strains in the community leads to an increase in mortality rates due to a reduction in the therapeutic options available. Thus, the development of effective therapeutics against a huge number of AR bacterial populations has become an urgent priority to combat antibiotic resistance. There are number of challenges involved in managing AR bacterial populations, including: i) the sequencing of culturable and non-culturable strains of bacteria; ii) the compilation of experimentally validated AR-associated information; iii) the prediction of AR-associated bacterial components; iv) the annotation of resistome or AR-associated genomes from whole-genome and meta-genome data; and v) the whole-genomes based prediction of antibiotic susceptibility. In the present era, whole-genome sequencing (WGS) of culturable and non-culturable bacteria is not a major challenge, but the compilation and mining of AR-associated information remains a major challenge.

Numerous databases, data-mining tools, and pipelines have been developed in the past two decades to investigate the mechanism of AR in bacteria and to predict bacterial AR genes and genomes. In this article, we critically review all AR-associated computational resources available to help the scientific community select the most appropriate resources to meet their needs. First, we review all of the databases that maintain direct or indirect AR-associated information. We classify these databases on the basis of content type and describe the utility and limitations of each resource. In order to help users to find the best resources, which are regularly updated, we recommend a few databases in

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each category. These AR databases are heavily used by computation tools to predict AR genes or genomes. Newly sequenced genes are searched against these databases using similarity search tools (i.e., BLAST) to identify AR-associated genes. In addition, machine-learning or deep-learning techniques have been used to predict AR genes in newly sequenced genomes. We recommend certain tools in each category, and provide detailed descriptions so that users can identify the most appropriate tool for their study. We have highlighted the type of information and queries taken up by each computational tool. *In silico* predictions and wide computational analysis have proven useful in increasing the speed of drug discovery [8,9]. This review also discusses pipelines, which provide a complete solution to the problem. In addition, we discuss alternative solutions to manage the treatment of AR-associated bacterial diseases. We hope that this information will assist researchers in addressing AR-linked problems and will aid in drug discovery.

Databases

The scientific community is actively collecting and compiling AR-related information obtained over the past two decades in order to gain an understanding of the mechanism of resistance. These resources are not being developed by any single group, university or country, but are the work of different independent individuals and groups. Thus, grouping these databases or resources is a tedious job as they contain overlapping content, and a number of older databases have been merged into new databases. Broadly, we classified these resources into four categories: i) databases of AR genes; ii) enzyme-specific databases; iii) organism-specific databases; and iv) other resources not covered in the first three categories (Table 1).

Databases of AR genes

Antibiotic Resistance Genes Online (ARGO) was the first database, developed in 2005, to list AR genes. It contains comprehensive information about 555 β-lactamase-resistance genes and 115-vancomycin resistance genes [10]. This database was never updated after 2005, and its site is not active. In 2007, the MvirDB database was been developed to catalogue AR genes as well as toxic proteins (toxins) and virulent proteins. This database contains limited AR genes and was also not updated after development; the site is no longer functional. A comprehensive database called the Antibiotic Resistance Genes Database (ARDB) was manually curated from the literature available in 2009. It was built to assist the characterization and identification of genes conferring antibiotic resistance [11]. ARDB is no longer maintained, but all the data can be found in a newer and up to date resource named The Comprehensive Antibiotic Resistance Database (CARD), which was developed in 2013. CARD contains molecular and sequence data for antibiotic resistance genes and their associated proteins and phenotypes [12,13]. It also contains information related to antibiotics and their targets, as well as the ontology of antibiotic resistance. This database is regularly updated and a number of papers have been published on the updated information. The updated release (Aug 8, 2020) contains 3057 reference sequences, 3103 antimicrobial (AMR) detection models and 1704 single nucleotide polymorphisms (SNPs). A number of prediction and analytical tools have been integrated into CARD, including: i) BLAST for analysing and annotating sequences; and ii) Resistance Gene Identifier (RGI) for predicting resistome genes. The most notable update in CARD 2020 is a new Resistomes & Variants module, which provides analyses and statistical summaries of in-silico-predicted resistance variants from 88 pathogens, 9560 chromosomes, 21,362 plasmids, 102,181 WGS assemblies and 222,011 alleles. The presence of analytical tools such as RGI, enhances the utility of CARD in predicting antibiotic resistance in genomic or metagenomic data.

There are also a number of resources or projects that maintain databases of AR genes. The National Center for Biotechnology Information (NCBI) BioProject compiles a wide range of biological data, and includes the regularly updated Bacterial Antibiotic Resistance Reference Gene Database (BARRGD). Another AR-related database maintained by the NCBI is the National Database of Antimicrobial Resistance Organisms (NDARO). This database contains a wide range of information on antimicrobial resistance genes, including their variants, classes, alleles, and gene families. This site integrates ARMFinderPlus, which allows users to find AR genes and resistance-associated point mutations.

β-lactamase-enzyme-specific database

One of the important enzymes responsible for degrading or modifying antibiotics is β-lactamase, which is produced in the periplasmic space. This enzyme hydrolyses the β-lactam ring of administered β-lactam antibiotics, rendering them harmless to bacteria [14]. β-lactam antibiotics are the broad-spectrum antibiotics most widely used for the treatment of severe Gram-negative infections. *Escherichia coli*, *Salmonella enterica* and *Klebsiella pneumoniae* are the most prevalent Gram-negative bacteria responsible for causing a variety of diseases in humans [1]. Thus, β-lactamases are considered to be the most important enzyme in the AR field, and numerous databases have been dedicated to these enzymes.

Historically, the first resource for β-lactamases was maintained by the Lahey clinic, who assigned new enzyme numbers to representatives of the β-lactamase family; the data were transferred to NCBI and are maintained in the BAARGD [15]. Among the active databases, the Lactamase Engineering Database (LacED), which contains information regarding TEM and SHV-β-lactamases, is the oldest, developed in 2009 [16]. The Metallo-β-Lactamase Engineering Database (MBLED), developed in 2012, contains information on class B β-lactamases [17]. Both LacED and MBLED are modules of the Lactamase Engineering Database, containing information about class A and B β-lactamases, respectively. Although both of these resources are still active, they are restricted to a particular class of β-lactamases. So, the Beta-Lactamase Database (BLAD) and the Comprehensive Beta-lactamase Molecular Annotation Resource (CBMAR) were developed in 2013–14 [18,19]. BLAD contains approximately 2000 gene sequences, as well as the 3D crystal structures of 200 β-lactamases and the physiochemical properties of their bound ligands. CBMAR provides information about the molecular and biochemical functionality of β-lactamases, classifying the enzymes on the basis of their hydrolytic character, sequences, antibiotic resistance, susceptibility to inhibitor, active

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TABLE 1

List of databases concerned with antibiotic-resistance (AR) genes.

Name and link	Description	Status	Update	Reference
ARGO http://bioinformatics.org/argo/beta/index.php	First database of AR genes	N	N/A	[10]
ARDB http://ardb.cbcb.umd.edu/	Information about AR genes	N	2009	[11]
CARD https://card.mcmaster.ca/	Comprehensive information on AR genes	A	2020	[12]
CARD 2017 https://card.mcmaster.ca/	Comprehensive information on AR genes	A	2020	
CARD 2020 https://card.mcmaster.ca/	Comprehensive information on AR genes	A	2020	[13]
BARRGD https://www.ncbi.nlm.nih.gov/bioproject/313047	Bacterial antibiotic resistance reference gene database.	A	Frequently updated	[15]
NDARO https://www.ncbi.nlm.nih.gov/pathogens/antimicrobial-resistance/	Antimicrobial resistance genes	A	2020	
LacED http://www.laced.uni-stuttgart.de/	Database of TEM-β-lactamases	A	2017	[16]
MBLED http://www.mbled.uni-stuttgart.de/	Contains class B β-lactamases	A	2012	[17]
BLAD	Widely circulated β-lactamases database	A	N/A	[18]
CBMAR http://proteininformatics.org/mkumar/lactamasedb/	Facilitates molecular annotation of β-lactamases	A	2014	[19]
BLDB http://bldb.eu/	Structural and functional information about β-lactamases	A	2020	[20]
β-lactamase database http://ifr48.timone.univ-mrs.fr/beta-lactamase/public/	Sequence, structure, function and phylogenetic tree of β-lactamases	A	N/A	
TBDReaMDB	Tuberculosis drug resistance associated mutation database	N	N/A	[21]
MUBII-TB-DB https://umr5558-bibiserv.univ-lyon1.fr/mubii/mubii-select.cgi	Database of mutations present in proteins and DNA of AR- associated genes of *Mycobacterium tuberculosis*	A	2013	[22]
u-CARE http://www.e-bioinformatics.net/ucare/	Comprehensive antibiotic resistance database of *Escherichia coli*	A	N/A	[23]
NPBAR https://www.pork.org/research/national-pork-board-antibiotic-resistance-database/	Research database to understand use of antibiotics on AR bacteria in pigs and humans	A	N/A	
Patient Safety Atlas	Contains a Health-associated Infections (HAI) progress report, AR data, antibiotic stewardship data and antibiotic use data	Replaced at Antibiotic Resistance & Patient Safety Portal (AR&PSP) (https://www.cdc.gov/hai/data/portal/AR-Patient-Safety-Portal.html)		
MEGARes https://megares.meglab.org/	Antimicrobial resistance database for population-level profiling	A	October 14, 2019	[24]
MEGARes 2.0 https://megares.meglab.org/	Antimicrobial resistance database for population-level profiling	A	October 14, 2019	[25]
EARS-Net http://data.europa.eu/euodp/data/dataset/antimicrobial-resistance-data	European Antimicrobial Resistance Surveillance Network containing data related to the occurrence and spread of AR bacteria in Europe	A	February 17, 2020	
Resistance Map https://resistancemap.cddep.org/AntibioticResistance.php	Contains maps and charts on the use of antimicrobials in the United States from 2000 to 2009	A	2020	
INTEGRALL http://integrall.bio.ua.pt/	Integrons, integrases and gene cassettes	A	2020	[28]
BacMet http://bacmet.biomedicine.gu.se/	Antibacterial Biocide & Metal Resistance Genes	A	2018	[31]
Mustard http://mgps.eu/Mustard/	Predicted AR genes in microbiota	A	2017	
Noradab Noradab.bi.up.ac.za	Non-redundant AR gene sequence	A	N/A	
ABRES finder	Consortium of AR genes prevalent in	A	N/A	

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TABLE 1 (*CONTINUED*)

Name and link	Description	Status	Update	Reference
http://scbt.sastra.edu/ABRES/index.php	India			
RAC	Repository of AR cassettes containing mobile genetic elements	N	N/A	[29]
PATRIC https://www.patricbrc.org/	Genomic-centric relational database	A	N/A	[33]
ResFams www.dantaslab.org/resfams	Database of protein families and associated profile hidden Markov models (HMMs)	A	2018	
FARME DB http://staff.washington.edu/jwallace/farme/	Functional AR elements	A	N/A	[27]
MvirDB	Database of AR genes, toxins and virulent protein.	N	N/A	[34]

A, available; N, not available.

site residues, fingerprints, mutations, location of genes, variants, phylogenetic tree, and so on.

The most comprehensive and updated database in the field of β-lactamases is the Beta-Lactamase Database (BLDB), developed in 2017 and continuously updated on a monthly basis [20]. It is a manually curated database, compiling sequence as well as biochemical and structural information for of all the currently known β-lactamases. The database is organized to include various classes of β-lactamases, their representative mutants, their respective kinetic profiles and their 3D structure. By November 10, 2020 it contained information about 4496 enzymes, 1269 structures, 167 mutants and 47 kinetic or hydrolytic profiles. Information such as 3D structure and kinetic profile is very useful in providing insight into structure–function relationships. We also came across an unpublished β-lactamase database, in which β-lactamases are organized according to the residues present on their active site, either serine in classes A, C and D or metal ion cofactors in class B. Each class is subdivided into various families, and each family has several variants that are linked with a protein description, accession number and localization. Moreover, structural information, kinetic information and minimum inhibitory concentration (MIC) data are also provided for each variant. Accession numbers are directly linked with the NCBI databases, and structural information is linked with the Protein Data Bank. Structural and kinetic information can provide an insight into the active-site residues of β-lactamases, so that their recognition and specificity with different antibiotics could be studied. Moreover, users can also develop MIC prediction models so that the MIC of certain antibiotics for any newly emerged β-lactamase variant could be predicted.

Organism-specific databases

TB is one of the deadliest infectious diseases, and the causal bacterium, *M. tuberculosis*, frequently acquires drug resistance through the mutation of specific genes. In order to provide comprehensive information, the TB Drug Resistance Mutation Database (TBDReaMDB) was developed in 2009 [21]. This database has vast utility in the development of new diagnostic tests that can rapidly identify the drug-sensitivity profiles of *M. tuberculosis* strains. However, mutation detection is required in order to use this database. Moreover, because of its loose syntax and grammar, the interrogation process within TBDReaMDB is a little complex. Therefore, the updated database MUBII-TB-DB was developed in 2014; this database contains a set of mutations present in the proteins and DNA of AR-associated *M. tuberculosis* genes [22]. The analysis of query sequence is easy and quick to perform using this database and does not require any prior bioinformatics training. This database can be used to identify *M. tuberculosis* mutations and possible antibiotic therapies. Moreover, it can be used for microbiological identification of other microbial mutations, and hence can be used for the surveillance and control of multi-drug-resistant microorganisms.

In order to study drug resistance in *Escherichia coli,* a manually curated database named User friendly Comprehensive Antibiotic resistance Repository of *Escherichia coli* (u-CARE) was developed in 2015 [23]. It contains 52 antibiotics, as well as around 107 genes plus SNPs and transcription factors that are involved in multi-drug resistance in *E. coli*. In addition, the National Pork Board antibiotic resistance database is a research database developed in 2017. Using risk-analysis-based categorization, this research project was designed to understand the effects of certain antibiotics on AR bacteria present in pigs and their environment, as well as the spread and dissemination of these bacteria to other pigs and/or humans.

Other databases

There are several other databases that deal with the AR field in some way. Resources like the Patient Safety Atlas (PSA), MEGARes, EARS-Net, and Resistance Map are largely based on phenotype, population level profiling and regions. Among these, the PSA is a web application that has four interactive datasets: AR data, Healthcare-Associated Infectious (HAI) data, outpatient antibiotic use data and inpatient antibiotic stewardship data. The AR data describes bacteria with resistant phenotypes such as CRE (carbapenem-resistant *Enterobacteriaceae*), MRSA (Methicillin-resistant *Staphylococcus aureus*) and multidrug-resistant *Pseudomonas aeruginosa*. The data also includes pathogen-antibiotic combinations that are used to define the resistance of the bacteria to a particular drug. The portal allows the visualization of regional, national and state-level data on antibiotic resistance. *Escherichia coli*, *Klebsiella* spp., and *Enterobacter* spp., belonging to Enterobacteriaceae, cause several infections including pneumonia, bloodstream infections, urinary tract infections, and so on. *Staphylococcus aureus* causes diseases

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including skin infections, bone infections, and heart-valve infections, whereas *P. aeruginosa* causes nosocomial infections, bacteraemia, and other infections. Moreover, emerging resistance in these pathogens makes their treatment very difficult. Hence, the PSA web application is a useful tool in studies related to resistant strains of this type. MEGARes, which was developed in 2017, is highly useful for population-level profiling of antimicrobial resistance [24]. It is also used for the development of high-throughput acyclic sequence classifiers and for hierarchical statistical analysis of large data. The latest version of this tool, MEGARes 2.0, was published in January 2020 [25]. In addition to the data provided in the previous version, it also incorporates metal-resistance determinants and biocide-resistant determinants. In addition, the European Antimicrobial Resistance Surveillance Network (EARS-Net) database contains data related to the occurrence and spread of AR bacteria in Europe, whereas Resistance Map contains maps and charts that contain data on the use of antimicrobials in the United States from 2000 to 2009, as well as data on antimicrobial resistance for the United States, Canada and more than 30 European countries for the year 2009.

ResFams and FARME DB contain the antibiotic resistance data derived from the metagenomes. ResFams is a database of protein families that have AR functions and their associated profile hidden Markov models (HMMs), which was curated in 2015. The database is structured and organized by ontology and thus explores the relationship between environmental and human-associated resistomes [26]. Similarly, the Functional Antibiotic Resistance Element Database (FARME DB), also known as the Metagenomic Element (FARME) database, contains DNA and protein sequences derived from metagenomic data and plays a role in conferring AR [27]. It also incorporates mobile genetic elements, regulatory elements and protein-flanking AR genes. Like ResFams, FARME DB also incorporates sequences from both clinical bacterial isolates and environmental bacterial isolates. However, the database also comprises information about regulatory and mobile genetic elements. The best-known resource dealing with mobile elements is INTEGRALL, developed in 2009, which provides detailed information about integrons, integrases, mobile elements called gene cassettes, DNA sequences and genetic arrangements [28]. INTEGRALL is an up-to-date database that is updated weekly, and it currently contains 11,469 entries for 1509 integrase genes and 8562 gene cassettes spanning across 213 genera and 465 species. Integrons are the primary mechanism for the acquisition of AR genes, and both integrases and gene cassettes are part of integrons. The data on this web-based platform will be helpful in revealing the role of integrons in bacterial interactions and adaptive responses. Similarly, the RAC (repository of antibiotic resistant cassettes) database, developed in 2011, contains a collection of gene cassettes, which are generally found integrated into integrons [29]. In addition, Multiple Antibiotic Resistance Annotator (MARA) is a database that contains mobile elements and mobile AR genes from Gram-negative bacteria, which was developed in April 2018 [30]. The accurate annotation in this database enables the comparative analysis of resistance genes and associated mobile elements in submitted sequences. Each sequence is indicated as containing potential composite transposons, truncated sequences and direct repeats (DR) from the list of orientations and positions of annotated features. Moreover sequences can be submitted to the site for annotation.

Beside antibiotics, antibacterial biocides and metals also contribute significantly to the development and maintenance of AR in bacterial communities by the virtue of co-selection. The Antibacterial Biocide & Metal Resistance Genes Database (BacMet), developed in 2014 and last updated on 11 March 2018, is a manually curated database of experimentally verified antibacterial biocide- and metal-resistance genes [31]. It actually consists of two databases; one is a manual curation of genes with experimentally confirmed resistance function and the other is a prediction of resistance genes based on sequence similarity. The major emphasis is on manually curated information on metal- and biocide-resistance genes, because biocides and metals can contribute to the development or maintenance of AR via co-selection. This database contains 753 experimentally confirmed and 155,512 predicted resistance genes, as well as 111 compounds including 58 antibacterial biocides and 23 metals.

The antibiotic resistance field is quite vast, and bioinformaticians have contributed significantly by providing several other miscellaneous resources such as ARGMiner, Mustard, Noradab, Antibiotic Resistance Gene Finder (ABRES finder), Pathosystems Resource Integration Centre (PATRIC) and MvirDB. ARGMiner is a web-based curation system that employs crowd sourcing and supports the annotation and inspection of several key attributes of potential AR genes, including their gene name, resistance mechanism, antibiotic category, mobility evidence and occurrence in clinically important bacterial strains [32]. Mustard is a database of AR determinants and curated gene sets identified in functional AR determinants. The database was developed in 2017 and contains 6095 AR determinants from 20 families in the human gut microbiota. Noradab, the Non-redundant antibiotic resistance database, which developed in 2018. It contains non-redundant antibiotic resistant gene sequences collected from the ARDB and CARD databases, but the webserver is sometimes unreliable and contains some errors, especially with links to other resources. Antibiotic Resistance Gene Finder (ABRES finder) is a consortium of AR genes that are prevalent in India. It contains a total of 37 antibiotics, 377 gene families and 36,467 genes. Similarly, PATRIC is a genomics-centric relational database developed in 2011 that is helpful to infectious-diseases researchers as it comprises all genome-scale data for pathogenic bacteria [33]. Another, MvirDB database was built in 2007 and integrates entries or information on the DNA and protein sequences of AR genes, protein toxins and virulence factors curated in other databases, which are hyperlinked to their original source [34]. Fig. 2 gives an overview of the timeline for various computational resources that can aid work on overcoming AR.

Prediction tools and software

Some of the databases discussed above also provide general AR gene prediction. For example, CARD includes tools for the analysis of molecular sequences, such as BLAST, as well as RGI software for resistome prediction on the basis of homology and SNP models. These resources are discussed in the above section,

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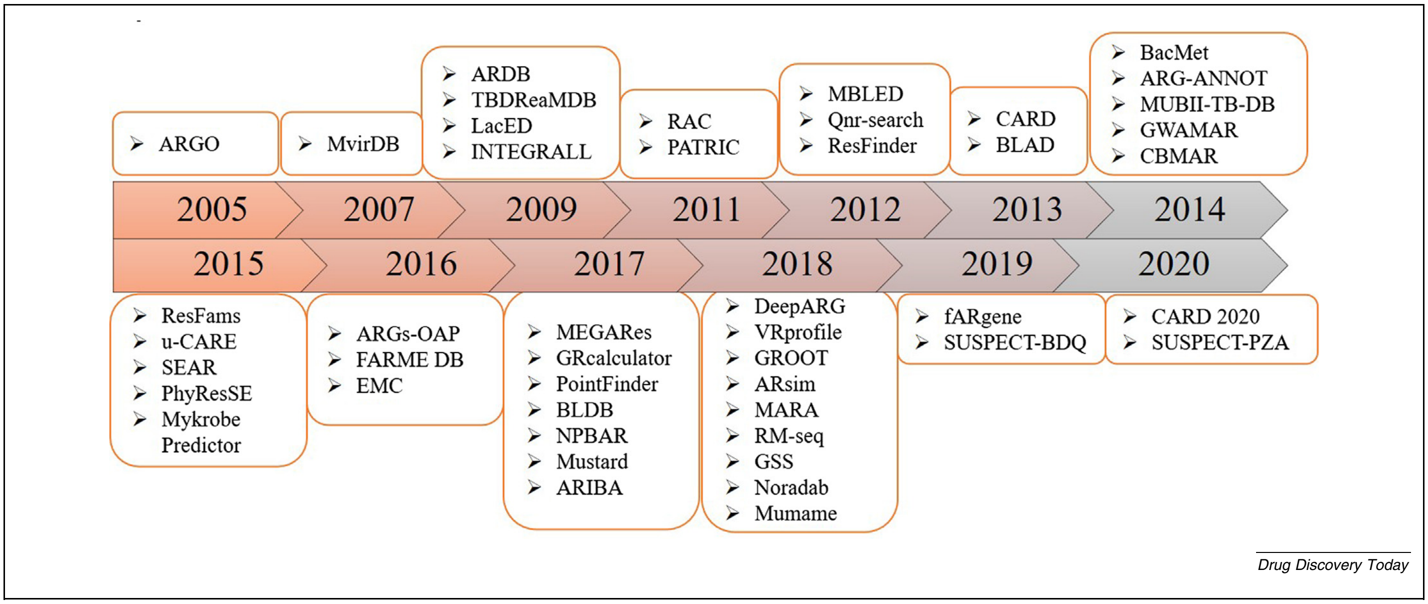

FIG. 2
Timeline of computational resources aiding in effort to combat antibiotic resistance.

so here, we provide insight into resources that have been developed solely for prediction purposes.

Prediction of AR genes

Antibiotic Resistance Gene-ANNOTation (ARG-ANNOT) is a bioinformatic tool for the detection of AR genes in bacterial genomes [35]. This tool was developed in 2014 and is based on BLAST. The updated version of ARG-ANNOT allows users to analyse sequence without using a web interface. In 2012, Boulund *et al*. [36] described a method to discover and annotate plasmid-mediated fluoroquinolone resistant (qnr) genes in fragmented nucleotide sequences. This tool can be used for the detection of existing resistant genetic determinants, point mutations that are known to be associated with AR, and putative new antibiotic resistance genetic determinants in bacterial genomes. DeepARG is a recent tool based on deep-learning methods that is used to predict AR genes with high precision of >0.97 and recall of >0.90 [37]. This tool can be used to predict AR genes in metagenomic data. The tool was developed in 2018 and is available both as a command line tool and as a webservice. DeepARG has utility in the metagenomic profiling of AR genes. The architecture of DeepARG is not restricted to AR genes and can be employed to train any set of genes, and hence it can be used to create new deep-learning model. Pairwise comparative modelling (PCM) is a 3D-based generic method to interpret resistance genes. It uses a homology model to increase the specificity of functional prediction for distantly related proteins. PCM uses a specific training approach to build structural models and to assess their relevance. It uses reference protein sequences of a given family, their structures and a series of negative references.

Mykrobe Predictor is a software package that analyses the complete genome of particular bacteria and predicts which antibiotic or drug the infection is resistant to within a couple of minutes. The software was developed in 2015 and is specially designed for *M. tuberculosis* and *S. aureus*. For *M. tuberculosis*, prediction has a sensitivity of 82.6% and a specificity of 98.5%, whereas for *S. aureus*, prediction has a sensitivity of 99.1% and a specificity of 99.6%. This tool has also proven that minor alleles can actually improve the detection of extremely drug-resistant strains [38]. Mykrobe Predictor can be used by microbiologists and doctors to choose the best treatment using the predictions of antibiotic or drug resistance. Moreover, species can be identified and information about resistance profile, virulence elements, and phylogenetic lineage can be obtained using this software. ARIBA is a tool developed in 2017 to help in the identification of AR-associated genes and SNPs directly from paired sequencing reads [39]. The developers claim that this tool is fast, efficient and accurate in comparison to current methods. To identify AMR genes and variants, ARIBA uses a combined mapping and targeted local assembly approach. The tool can be used to obtain a deeper and better understanding of the resistance associated with each isolate because it delivers significantly more results than the existing tools, especially for variant cells. ABRicate is another tool developed to find AR genes. It performs mass screening of contigs for the identification of AR and virulence genes using a Perl-based program. Table 2 lists all of the computational tools used to assist in characterizing AR.

Prediction of AR-associated mutations

ResFinder is an up-to-date (last updated in 2019) tool developed in 2012 for the identification of acquired AR genes and chromosomal mutations in whole-genome data. The method was evaluated on 1411 different resistance genes and 23 *de-novo*-sequenced isolates [40]. The fourth version of the tool can be accessed at https://cge.cbs.dtu.dk/services/ResFinder/. ResFinder can also be used to detect antimicrobial resistance genes in metagenomic samples from permafrost [41]. Another mutation-detecting tool called Genome-wide assessment of mutations associated with drug resistance in bacteria (GWAMAR) was developed in 2014 to detect drug-resistance associated mutations by comparative

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TABLE 2

List of prediction tools and software available for research on antibiotic resistance.

Name and link	Description	Update	Reference
ARG-ANNOT http://backup.mediterranee-infection.com/article.php?laref = 282&titre = arg-annot	Tool to detect AR genes in bacterial genomes	2018	[35]
ARGMiner http://bench.cs.vt.edu/argminer	Annotation and prediction of ARG	2020	[32]
Qnr-search http://bioinformatics.math.chalmers.se/qnr/	Method to discover and annotate plasmid-mediated fluoroquinolone resistant (qnr) genes	N/A	[36]
DeepARG http://bench.cs.vt.edu/deeparg	Predicts AR genes in metagenomic data	2019	[37]
PCM https://github.com/aghozlane/pcm	3D-based generic method to interpret resistance genes	2018	
Mykrobe Predictor https://www.mykrobe.com/	Predicts which antibiotic or drug the infection is resistant to	2019	
ARIBA https://github.com/sanger-pathogens/ariba	Tool for the identification of antimicrobial-resistance-associated genes and SNPs	2019	[39]
ABRicate https://github.com/tseemann/abricate	Tool for identification of antimicrobial-resistance and virulence genes	2019	
ResFinder 4.0 https://cge.cbs.dtu.dk/services/ResFinder/	Tool to identify acquired AR genes and chromosomal mutations in whole-genome data	August 2019	[41]
GWAMAR http://bioputer.mimuw.edu.pl/gwamar/	Tool to detect drug-resistance-associated mutations	December 2015	[42]
Excess mutational codes https://github.com/kassonlab/positionalmi	To predict local and allosteric mutations that are affecting β-lactamase drug resistance	N/A	[43]
RM-seq https://github.com/rguerillot/RM-seq	A workflow for the detection and functional estimation of mutational resistance in bacterial population	January 2019	[44]
PointFinder https://cge.cbs.dtu.dk/services/ResFinder-3.0/	Detects antimicrobial-resistance-associated with chromosomal point mutations	2019	[45]
VRprofile http://bioinfo-mml.sjtu.edu.cn/VRprofile/	Tool that helps in the exploration of virulence and AR genes	2016	[48]
GROOT https://github.com/will-rowe/groot	Tool for resistome profiling	2019	
PhyResSE http://phyresse.org	Tool describing AR in *Mycobacterium tuberculosis* and other lineages from whole-genome sequencing data	N/A	[50]
GRcalculator http://www.grcalculator.org/grtutorial/Home.html	Growth rate calculator used to carry out drug–response studies	N/A	[51]
ARsim	Simulation tool for the implementation of bacterial growth and AR	N/A	[52]
SUSPECT-PZA http://biosig.unimelb.edu.au/suspect_pza/	Identification and analysis of resistance mutations in the pyrazinamidase protein of *M. tuberculosis*	February 2020	
SUSPECT-BDQ http://biosig.unimelb.edu.au/suspect_bdq/	Identification and analysis of resistance mutations in the ATP synthase C polypeptide of *M. tuberculosis*	2019	[47]
fARGene https://github.com/fannyhb/fargene	Identification and reconstruction of AR genes in metagenomic data	August 2019	[53]
Mumame http://microbiology.se/software/mumame	Quantification of gene-specific point-mutations in metagenomic data	October 2018	

analysis of the whole-genome sequences of bacteria. First, GWAMAR employs eCAMBer to identify homologous gene families. Second, it identifies mutations on the basis of multiple alignments in the strains of interest. Third, it employs statistical methods to identify which mutations are most associated with drug resistance. Using the tool, a set of novel putative drug-resistance-associated mutations was identified from datasets of *M. tuberculosis* [42]. In 2016, Cortina *et al.* [43] wrote a tool named Excess mutational codes, which is freely available on github. Their code uses positional mutual information in molecular dynamics simulation to predict local and allosteric mutations that affect β-lactamase drug resistance. The utility of this tool was demonstrated by its ability to predict residues linked to the catalytic activity of CTX-M−9 β-lactamase; six of the eight predicted mutations were associated with reduced resistance towards cefotaxime of more than two-fold (the CTX-M family of bacteria show resistance towards cefotaxime antibiotic) [43].

The ability to detect both local and allosteric amino acid residues linked to the catalytic activity of the β-lactamase enzyme can be used in to predict newer β-lactamase variants that have a role in bacterial drug resistance. This prediction can be very helpful for clinical and epidemiological surveillance. Recently, in 2018, Guérillot *et al.* [44] developed RM-seq, a deep-sequencing workflow based on the molecular-barcoding method. The tool is helpful in the detection and functional estimation of mutational resistance in a bacterial population. The detection of very-low-frequency bacterial resistance sub-populations allows the discovery of rare resistant bacterial populations and the characterization of antibiotic-associated mutations. Moreover, phenotypic screening of mutations can be carried out if an

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accurate quantification of resistance mutations is obtained using the technique [44]. This deeper understanding can serve as the basis of optimised antibiotic-treatment regimens, which can help in treating microbial infections. PointFinder was developed in 2017 for the detection of antimicrobial resistance associated with chromosomal point mutations in pathogenic bacteria [45]. The web-based PointFinder service is available as a feature in ResFinder (https://cge.cbs.dtu.dk/services/ResFinder-3.0/), whereas a python script is available at https://bitbucket.org/genomicepidemiology/pointfinder/src/master/. PointFinder can report detected mutations along with nucleotide and amino acid changes. Mutations conferring resistance to quinolones, macrolides and polymyxin in *E. coli*, *Salmonella* and *Campylobacter jejuni* can be obtained from PointFinder.

In addition, there are tools that utilize the molecular dynamic technique to identify mutations. Pyrazinamide (PZA) is a drug that is used for the treatment of tuberculosis, but resistance to pyrazinamide sometimes occurs due to a mutation in the pyrazinamidase protein (PncA) of M. tuberculosis. Using the structural information for PncA, a user-friendly web interface, StrUctural Susceptibility PrEdiCTion for pyrazinamide (SUSPECT-PZA), was developed in February 2020 to identify and analyse resistance mutations in PncA. The structural insights that are provided by this tool can help in making decisions regarding the patient's treatment [46]. Similarly, another anti-tuberculosis compound, bedaquiline (BDQ), became resistant due to mutations in AtpE (ATP synthase C polypeptide). Using structural information on AtpE, another user-friendly web interface, StrUctural Susceptibility PrEdiCTion for bedaquiline (SUSPECT-BDQ) was developed to predict missense mutations and to provide structural insight into AtpE [47].

Tools for the genome-level annotation of AR

VRprofile is a web-based service, developed in 2018, to assist in the exploration of virulence and AR genes. It co-localizes the homologs of conserved gene clusters using BLASTp or HMMer searches and performs better for island-like region predictions than other methods. This tool can be used to aid the real-time definition of disease-relevant gene clusters in pathogenic bacteria and also to meet the increasing demands of the re-annotation of variable regions of bacterial genomes [48]. Resistome profiling is a complicated process due to similarity with reference genes, sheer volume of sequencing data and complex analysis workflows, so Graphing Resistance Out Of meTagenomes (GROOT) was developed in November 2018 to profile the resistome in an efficient and accurate way. This method also has the capacity to improve current metagenomic workflows. To carry out fast classification of metagenomic sequence reads using similarity-search queries, the method uses indexed variation graph representation of gene sets with a locality-sensitive hashing forest indexing scheme [49], so the tool can be used to sort resistance genes in metagenomic samples. PhyResSE is another web tool developed in 2015 to describe AR in *M. tuberculosis* and other lineages from whole-genome sequencing data. The tool provides a high-tech methodology to untrained physicians and scientists. Elaborate data processing and quality control tools are combined in PhyResSE in a way that suits human diagnostics [50]. The web interface can also be used for the identification of multidrug-resistant *M. tuberculosis* complex (MTBC)-resistance-mediating variants and the classification of phylogenetic lineages.

GRcalculator, which was developed in 2017, is a growth rate calculator used to improve the value and reliability of drug–response studies in clinical research, as such studies are complicated by differences in cellular growth across diverse samples. Drug–response data are calculated, analysed and visualised using this approach. The GRmetrics R package is also available for offline data analysis and visualization (https://github.com/datarail/gr_metrics). GRmetrics is much better than IC_{50} and Emax for measuring cellular response to drugs. GRcalculator is a user-friendly and powerful tool that generates publication-ready figures [51]. The tool can be helpful for users in academia as well as in industry and can be used to carry out reliable and reproducible comparisons of drug efficacy and potency across cell lines that have different cell division rates.

In addition, the ARsim simulation tool was developed in 2018 to model bacterial growth and antibiotic resistance in order to decide the suitability of specific antibiotics for use against AR bacteria. It also incorporates vertical and horizontal AR gene transfer mechanisms. The tool was developed with the intention of providing researchers with an effective method to look for new ways to fight antibiotic resistance [52]. Bacterial growth can be simulated using the tool by growing bacteria in a virtual environment with predefined parameters. The effect of the addition of antibiotics to different types of bacteria can also be predicted.

Metagenome-based tools

Fragmented Antibiotic Resistance Gene iENntifiEr (fARGene) is a method or tool developed in April 2019 that can be used to identify and reconstruct AR genes directly from shotgun metagenomic data (fragmented or longer sequences) [53]. It is an assembly-based platform that provides both the nucleotide and the amino acid sequences of the reconstructed AR genes as output. It contains a number of developed and optimized models of resistance gene types. Moreover, the tool enables the user to create and optimize models for their choice of resistance genes. However, if the users are interested in quantifying gene-specific point-mutations in shotgun metagenomic data, then they can use a tool called Mumame, which was developed in September 2018. Mumame has the capability to differentiate between wild-type and mutated sequences in metagenomic data and to calculate their relative abundance. Functionally relevant mutations of even evolutionary distant homologs can be detected by this software tool. Therefore, the method does not essentially require well-characterized species and can work on complex metagenomes.

Pipelines and surveys

Related to AR genes

Search Engine for Antimicrobial Resistance (SEAR) is a web interface and pipeline built in 2015 in order to detect horizontally acquired antimicrobial resistance genes from raw sequence data. The pipeline provides information about resistance genes, estimation of their abundance and their reconstructed sequence, and links to additional information on each gene. The pipeline works by utilizing clustering and mapping to annotate full-length genes and their local alignment. The web tool is effective

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TABLE 3

List of pipelines and surveys available for research on antibiotic resistance.

Name and link	Description	Update	Reference
SEAR	Pipeline built to detect horizontally acquired antimicrobial resistance genes from raw sequence data	2018	
ARGs-OAP https://smile.hku.hk/SARGs	Tool to classify and annotate AR-gene-like sequences from metagenomics data	2019	[55]
Gonorrhoea surveillance study https://pathogen.watch/	Genomic survey of multidrug-resistant clones of *Neisseria gonorrhoea*	2019	[56]
Pfizer's ATLAS https://atlas-surveillance.com/	The Antimicrobial Testing Leadership and Surveillance database	2020	

in the detection of AR genes in sequencing data from clinical isolates and in environmental metagenomics data [54] (Table 3). Antibiotic Resistance Genes — Online Analysis Pipeline (ARGs-OAP) was built to classify and annotate ARG-like sequences from metagenomics data. ARGs-OAP was developed in 2016 after first constructing the Structured ARG reference database (SARG) using the ARDB and CARD databases, removing redundant sequences and optimizing query sequence identification by similarity. Hence, ARGs-OAP was developed using SARG and earlier hybrid functional gene annotation pipelines [55], and it can be used to conduct the metagenomic analysis of ARGs efficiently. This can help in reducing the dissemination of ARG to the environment and to the pathogens.

Miscellaneous

The gonorrhoea surveillance study, carried out in 2018, is a genomic survey of multidrug-resistant clones of *Neisseria gonorrhoea*. This work included the development of a web platform (https://pathogen.watch/) in order to predict antimicrobial resistance, molecular typing, multilocus sequence typing, multi-antigen sequence typing and phylogenetic clustering in combination with phenotypic and epidemiological data [56]. Genomic mapping of emerging antibiotic resistance in *N. gonorrhoea* strains that are responsible for causing sexually transmitted disease can help researchers and doctors to monitor and prescribe the most effective antibiotics. Pfizer's ATLAS (The Antimicrobial Testing Leadership and Surveillance) database integrates three surveillance programs — AWARE, TEST and INFORM — and proved to be a key tool in the fight against AR genes. This tool is helpful in providing researchers and healthcare practitioners with information about resistance trends in their areas. It is available as a webservice and as a mobile application (Table 3). The database can help to monitor real-time changes in pathogen resistance and to detect trends in multi-drug resistance longitudinally over time.

Case study: How to obtain updated information about a recent resistant marker

The property of resistance in β-lactamase enzymes is encoded by genetic sequences called resistance markers. β-lactamase-producing bacteria cause polymicrobial infections [57]. They either have direct pathogenic impact or indirect impact through the production of β-lactamase enzymes. The infections caused by these bacteria could be any of the infections caused by Gram-negative bacteria, such as urinary tract infections, pneumonia, wound infections, blood stream infections, surgical site infections, respiratory tract infections, skin infections, soft tissue infections, sexually transmitted diseases, plague, tuberculosis and so on. These infections can lead to serious life-threatening complications such as kidney failure, sepsis, toxic shock syndrome and even death.

Patients who do not recover from one of these infections after administration of antibiotics are tested for antibiotic resistance. After extracting DNA from a patient's blood sample, the DNA is sequenced. The quality processed read obtained in this way can be mapped against gene and protein sequences from databases of AR genes. Suppose the identified gene shows high similarity with one of the NDM (New Delhi Metallo-β-lactamase) β-lactamase genes. (Bacterial isolates carrying the NDM gene have the capability to destroy penicillin, ampicillin, cephalosporin and every other β-lactam antibiotic.) Then this newly identified gene could be a new NDM variant. The researcher could think of building a platform to predict the resistance and susceptibility of NDM, or any related β-lactamase, to various antibiotics and drug molecules. For that, the researcher will need databases containing information about nucleotide sequences, protein sequences, structures and other characters of NDM genes. For example, computational resources such as the Beta-Lactamase database (BLDB) provides information about all of the β-lactamases discovered to date [20]. Using this resource, we can see that β-lactamases can be divided into four classes (Classes A–D) on the basis of their sequence identities. Class A has 93 resistant markers, class B has 86, class C has 38 and class D has 12 resistance markers. Each marker has several variants in bacterial populations: 93 resistance markers of class A have 1416 variants, class B has 635 variants, class C has 1360 variants and class D has 933 variants. These markers and their variants are either naturally occurring or are acquired. Information about their gene sequence, protein sequence, the organism in which they were first isolated, their structures, and so on can also be extracted using this computational resource. Class B β-lactamases, containing 86 markers, are subdivided into three sub-classes: class B1, B2 and B3. The NDM resistance marker, which has 27 variants discovered to date, is present in class B1 β-lactamases. Moreover, information about the MIC or the resistance/susceptibility profile of various antibiotics must also be included in the databases. After building datasets containing this information, prediction models can be constructed to aid in predicting the resistance/susceptibility profile of any new NDM variant. Similarly, using several HMM-based methods, new AR genes can be detected from shotgun sequencing data. This will help in providing an under-

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standing of novel resistance factors that might be encountered in clinics in the near future.

Discussion and conclusions

Antibiotic resistance is continuously increasing and is a major global threat. According to the WHO, drug-resistant diseases could cause 10 million deaths each year by 2050, and by 2030, AR could force up to 24 million people into extreme poverty. Currently, at least 700,000 people die each year due to drug-resistant diseases [58]. Even in the current global pandemic, COVID-19, AR plays a significant role and complicates the disease course of some patients. A noteworthy report says that 10% of COVID-19-affected patients developed secondary bacterial infections [59]. In some cases, these patients harboured antibiotic-resistant A. baumannii, which can cause septic shock, resulting in severe organ damage and in some cases death [60]. Several studies suggest that nearly all COVID-19 patients are given antibiotics as part of their treatment regimen [61]. Such widespread use of antibiotics is also playing a major role in driving the evolution of antibiotic-resistant bacteria. Hence, gaining an insight into the expanse of the resistome and its transfer to other bacteria is crucial for controlling the spread of AR genes [5]. The advent of WGS technologies has significantly influenced science, including AR research as it provides the ability to predict AMR by analysing next-generation sequencing (NGS) data. Various studies have predicted the AMR and even the MIC of pathogens such as N. gonorrhoea, *Salmonella*, K. pneumoniae, and P. aeruginosa, by employing machine-learning or deep-learning techniques to sequencing data [62–68]. Similarly, the microbiome and antibiotic resistome have been characterized by analysing metagenomics sequencing data [69–74].

For decades, antibiotics have remained a major saviour in combating several infectious diseases, but the emergence of resistance has shifted the paradigm towards peptide therapeutics [75–77]. For years, peptides have been studied for potential antimicrobial activities [78,79], and a number of peptides (compiled in AntiTbPdb) have been found to be effective against mono-resistance, poly-resistance and even MDR-TB [80]. Several studies have shown that a combination of cell-penetrating peptides and antibiotics can be effective against several resistant pathogens [81–83]. In the past, various tools, such as AntiTbPred and AtbPpred, have been developed to predict peptides with different therapeutic properties that could be utilized in the design of effective antitubercular peptides [84,85]. Further examples include AntiFP for antifungal peptides [86], AntiMPmod for modified antimicrobial peptides [87], and CellPPDmod, MLCPP and KELM-CPPpred for cell-penetrating peptides [88–90]. In addition, vaccination is a promising strategy to protect from several infectious diseases. A lot of *in silico* tools have been developed to design immunotherapeutic and peptide-based vaccines [8,9]. VacTarBac (http://webs.iiitd.edu.in/raghava/vactarbac/) encompasses several epitope-based vaccine candidates against 14 major pathogenic bacteria [91]. Nevertheless, the half-life of peptides remains a major concern, which restrains them from overcoming antibiotics.

Databases containing information about AR genes, proteins, enzymes, and so on, as well as novel tools to obtain insights into AR from high-throughput sequencing data, promise to make significant contributions to reducing the economic and human burdens of antimicrobial-resistant disease-causing organisms. The resources discussed in this review help in facilitating preclinical research in the drug development process. While compiling this review article, we came across a number of AR-gene-related databases, most of which were published 5–10 years ago and updated only infrequently, if at all, since publication. The ARDB database was published in 2009 [11], and its web address (http://ardb.cbcb.umd.edu/) mentions that it is no longer being maintained, and that all the data can be found in the newer and up-to-date CARD resource. Other databases lack this kind of clarity. In addition, several databases, such as the Resqu, MARA and National Pork Board antibiotic resistance databases, are commercial and have restricted access that excludes many healthcare professionals. In our opinion, the most comprehensive and up-to-date database resources suitable for providing information about AR genes and their classes, alleles, gene families, ontology, encoded proteins and nucleotide sequences, as well as about AR drug targets, are CARD and NDARO. INTEGRALL is also an up-to-date database providing information about integron's DNA sequences and genetic arrangements. BLDB is most suitable for structural and functional information about β-lactamases. Researchers have developed tools in the past, such as ARG-ANNOT, which can detect existing and putative new AR genes using the local BLAST program in Bio-Edit software, but the dataset has not been updated for more than a year. DeepARG is more significant for the prediction of AR genes in metagenomic data, and it has the advantage of offering command line and web interface access.

Perspective

While extensively searching the literature, we came across several updated databases. Still, somehow the field is quite open to formulating new algorithms and software, which can aid in combating antibiotic resistance. Each pathogen exhibits a specific resistance mechanism and evolves with time and use of antibiotics. These complex resistance mechanisms make it difficult to accurately predict phenotypic antimicrobial resistance from genomic data, because the expression of genes depends on several factors. Researchers have applied various machine-learning and deep-learning techniques, but they tend to be specialized to specific bacteria. Besides this, each machine-learning technique has its flaws, and new algorithms have been developing. As more comprehensive up-to-date databases are developed, these techniques will perform better and will provide more accurate phenotypic antibiotic resistance predictions. Furthermore, robust models to predict the spread of resistance genes at local and global levels are needed. Newer algorithms must include the prediction of geographic location in addition to gene sequence, mechanism, and so on. Sequencing cost has decreased appreciably in the past, but it still surpasses many healthcare budgets in developing countries. As it becomes more affordable, metagenomics will be used more commonly to explore the potential role of different environments in the ecology of resistance. We hope that the accurate and rapid identification of AR genes will augment sequence-based personalized medicine.

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Conflict of interest

The authors declare no competing financial interests.

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